

Technology Stack

Date	15 July 2026
Team ID	XXXXXX
Project Name	FloodGuard - AI/ML Flood Prediction System
Maximum Marks	2 Marks

Technology Stack Details

S.No	Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3 (Flexbox/Grid), Vanilla JavaScript	Lightweight, no build step needed; AJAX (fetch) drives dynamic results without page reload
2	Backend / Server-Side	Python 3.12, Flask 3.1 (application-factory pattern)	Simple, well-documented framework; matches the team's Python/ML skillset
3	Database / Data Storage	SQLite via Flask-SQLAlchemy	Zero-config relational DB, sufficient for prediction logging at this scale
4	Machine Learning	scikit-learn, XGBoost, pandas, NumPy, joblib	Industry-standard libraries for classification, preprocessing and model persistence
5	Data Visualization	Matplotlib, Seaborn	Used in train.py to auto-generate EDA and model-comparison charts
6	Cloud / Hosting / Deployment	Render (Web Service, free tier), Gunicorn WSGI server	Deploys directly from GitHub with zero DevOps overhead; Gunicorn is the standard production server for Flask
7	Version Control & CI/CD	Git, GitHub (auto-deploy on push to main)	Every push to main triggers an automatic Render rebuild and redeploy